# A New Method of Video SAR Ground Moving Target Detection and Tracking Based on the Interframe Amplitude Temporal Curve

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***Abstract*— In the video synthetic aperture radar (Video SAR) system, the moving target will leave a shadow at its actual position due to the Doppler effect. As the shadow of the moving target moves between Video SAR frames, the amplitudes of pixel points at the corresponding positions will jump between frames as well. According to this characteristic, a new method of Video SAR ground moving target detection and tracking based on the interframe amplitude temporal curves is proposed in this article. In this method, the specially designed multiple receptive field fusion neural network model based on frame variation (MRFN-FV) is used to classify the pixel points with obvious interframe amplitude jumps on the whole-time axis, and then, the false alarms are suppressed based on the temporal change characteristics of pixel points. Finally, the improved clustering algorithm is used to detect, locate, and track the moving targets in each frame of SAR images. The effectiveness of the proposed method is verified through the measured data recorded by the THz band Video SAR system.**

***Index Terms*— Clustering algorithm, ground moving target detection, multiple receptive field fusion neural network model based on frame variation (MRFN-FV), video synthetic aperture radar (Video SAR).**

## I. Introduction

SYNTHETIC aperture radar (SAR) is a high-resolution and active microwave Earth observation system, which can obtain high-resolution radar images under extremely harsh meteorological conditions [1], [2], [3], [4], [5]. On the basis of SAR imaging and with the help of the SAR-ground moving target indication (GMTI) model, moving targets in the SAR echo can be detected. However, the traditional SAR-GMTI model can only obtain information about the moving targets within the synthetic aperture time and cannot continuously track the moving targets, so it is difficult to meet the practical applications.

To solve the above problems, Sandia National Laboratories (SNL) proposed the concept of Video SAR and realized high-frame-rate imaging of the region of interest (ROI) on the airborne platform (the imaging frame rate is higher than 5 Hz) [6], [7], [8]. Compared with the traditional SAR system, the Video SAR system can obtain dozens or even hundreds of continuous temporal SAR images in ROI, so it can detect and track the moving targets more accurately in ROI [9], [10], [11], [12]. Video SAR, which features high resolution and high frame rate, can replace optical or infrared devices in severe weather conditions to continuously monitor ROI.

Meanwhile, Video SAR moving target detection technology has gradually attracted the attention of many scholars. The constant false alarm rate (CFAR) detector [13] is frequently used to detect moving targets after clutter suppression. After the clutter suppression of Video SAR images with complex backgrounds, a large amount of residual clutter is often left, and especially, CFAR detection under the urban background will produce a lot of false alarms. If traditional clutter suppression methods (such as single-channel frequency-domain filtering algorithm, multichannel displaced phase center antenna (DPCA), along track interferometry (ATI), and space–time adaptive processing (STAP) algorithms [14], [15], [16]) are directly applied to the Video SAR-GMTI model, the advantages of Video SAR will not be fully exploited. In order to make full use of the characteristics of Video SAR interframe changes and the features of moving targets in the scene, coherent change detection and track-before-detection (TBD) were proposed for Video SAR moving target detection [17], [18]. On this basis, Zhang et al. [19] proposed a threshold segmentation algorithm based on maximizing Tsallis entropy to separate the moving target from the background and then used the three-frame difference method to detect the Video SAR moving targets. The above methods are used to detect moving targets based on the adjacent interframe information, and their algorithms depend on the accuracy of image registration and are sensitive to the clutter in the Video SAR images. The fluctuation of clutter in a complex background makes the detection results of moving targets unsatisfactory. In [20], the background of Video SAR was considered as a low-rank matrix and the moving target

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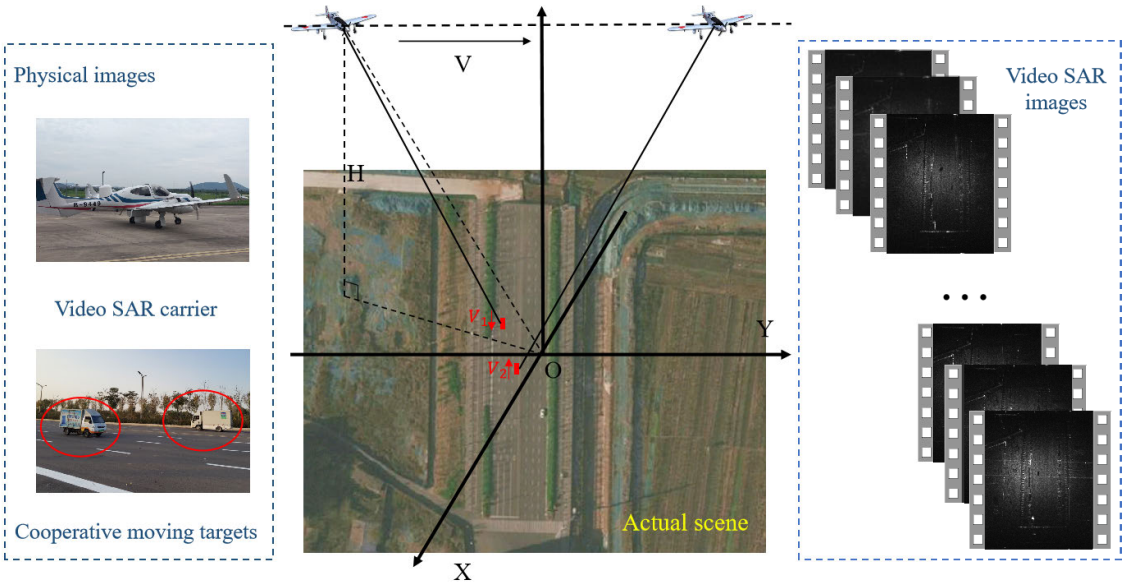



Fig. 1. Field experiment figures of the real video SAR system.

component as a sparse matrix, and a method of Video SAR moving target detection and tracking was proposed based on the separation of low-rank and sparse matrix, which was verified through simulation and measured data.

With the development of deep learning [21], a neural network is widely applied in the target recognition and detection of optical images and videos. Compared with optical images, the shapes of moving targets in the Video SAR images are relatively simple. Prof. Ding first introduced the deep learning theory to Video SAR moving target detection and tracking in [22] and further proposed related improvement methods in [23] and [24]. The effectiveness of the related methods was verified through simulation data and measured data. However, in practical applications, Video SAR moving target detection based on deep learning still needs to solve the problems of few moving target samples and the poor generalization ability of neural networks in different SAR backgrounds.

As the Doppler effect of the Video SAR system is rather significant, the moving target component in the Video SAR image will shift along the azimuth direction and then leave a shadow at the position of the moving target [25]. The motion of the shadow of the moving target between Video SAR image frames will cause jumps in the amplitudes of pixel points at relevant positions on the time axis. Based on the above characteristics, a designed neural network is proposed in this article to detect the pixel points with obvious interframe amplitude jumps on the whole-time axis, thus realizing the detection and tracking of the Video SAR moving targets.

The innovation points of this article are given as follows.

1) After Video SAR data are preprocessed, it is proposed that pixel points in the Video SAR scene can be divided into three categories: pixel points where the moving target appears (appear), pixel points where the moving target leaves (leave), and background pixel points (background). MRFN-FV, a classified neural network, is specially designed to detect the pixel points with obvious interframe amplitude jumps on the whole-time axis. This neural network can filter the false alarms caused by the amplitude changes brought by radar cross section (RCS) fluctuations of the same target at different viewing angles of Video SAR. Compared with the neural network based on the morphological features (2-D images) of Video SAR moving targets, the designed neural network is mainly used to detect the pixel points (1-D curve) with obvious interframe amplitude jumps on the whole-time axis, easily obtaining a large number of training samples in practical applications and having good adaptability to different Video SAR scenes.
2) Based on the above MRFN-FV neural network, a complete set of methods for Video SAR moving target detection and tracking are designed, including the false alarm suppression method designed by using the temporal change differences between the pixel points of moving targets and other pixel points in the scene, and steps for high-precision location and tracking of Video SAR moving targets by using the amplitude-based density clustering algorithm. The proposed method is verified by combining with the measured data of a real Video SAR system, and it is promoted in the actual Video SAR system.

## II. Introduction of Video SAR System

### A. Practical Experiment of Real Video SAR System

In the experiment, multiple frames of images in the same scene were obtained by the Video SAR system operating in the spotlight mode, as shown in Fig. 1. The optical images of the airborne platform and two cooperative moving targets are given on the left-hand side of Fig. 1. The Google Earth satellite image of the experimental scenario and the spatial geometric relationship diagram of the experimental process are displayed in the middle of Fig. 1. The multiframe Video SAR imaging results produced in this experiment are presented on the right-hand side of Fig. 1. It can be seen from the

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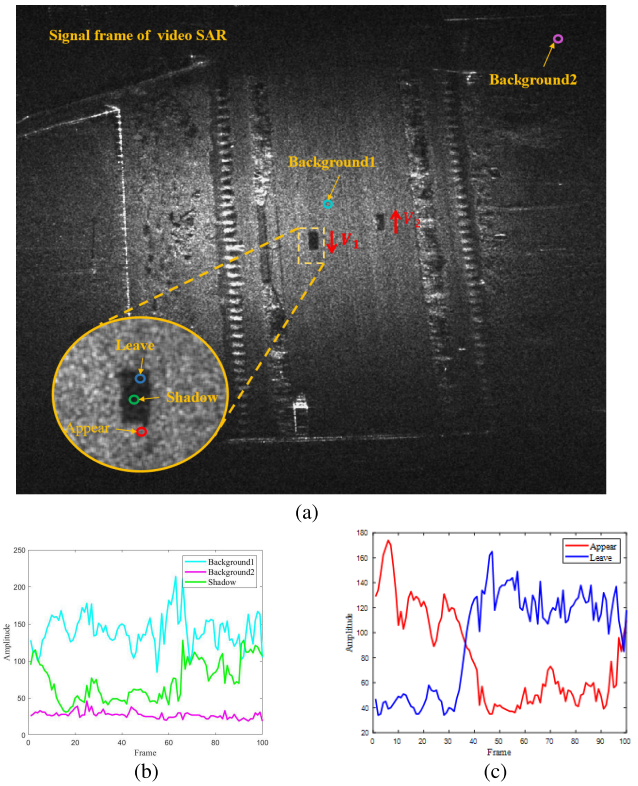



Fig. 2. Amplitude temporal curves of pixel points at different positions in the Video SAR scene. (a) Schematic of classification of pixel points in the Video SAR scene. (b) and (c) Amplitude temporal curves of different pixel points.

Video SAR images that there are obvious shadow features at the positions where the two moving targets are located. The method of Video SAR moving target detection and tracking proposed in this article is verified through the measured data recorded by the system.

### *B. Characteristics of Moving Targets Provided by Interframe Information*

As the Video SAR system works in a THz band, the Doppler effect will be more significant, and the tiny speed of moving targets will produce a Doppler shift. This makes the moving target deviate from its position and leaves a shadow at the actual position. As the moving target moves, the position of its shadow will change as well, which will cause a temporal change in the interframe amplitudes of pixel points. The moving targets can be indirectly detected and located by detecting the pixel points with obvious interframe amplitude jumps on the whole-time axis.

However, the echoes in the actual Video SAR scene are relatively complex, and there exist various nonideal factors. The following is a detailed analysis based on the measured Video SAR data, as shown in Fig. 2. Through the analysis of the measured Video SAR data, the change of amplitude temporal curves of pixel points in Video SAR images can be divided into the following situations.

*1) Situation 1:* During the observation time, the moving target leaves a pixel point. For example, the pixel point at the position of the blue circle in Fig. 2(a) changes from the shadow state of the moving target to the SAR background state. The amplitude temporal curves of pixel points will obviously go up during the observation time, as shown by the blue curve in Fig. 2(c).

*2) Situation 2:* During the observation time, the moving target appears at a pixel point. For example, the pixel point at the position of the red circle in Fig. 2(a) changes from the SAR background state to the shadow state of the moving target. The amplitude temporal curves of pixel points will obviously go down during the observation time, as shown by the red curve in Fig. 2(c).

*3) Situation 3:* During the observation time, the pixel point at a certain position is always in the shadow of the moving target. For example, the pixel point at the position of the green circle is shown in Fig. 2(a). As the moving target is too long or the moving speed is too slow, the pixel points near the middle position of the moving target will always be in a shadow state during the observation time. The amplitude temporal curve of the pixel point is shown by the green curve in Fig. 2(b).

*4) Situation 4:* During the observation time, the pixel point at a fixed position is always in the SAR background state. For example, the pixel points at the positions of the purple circle and the cyan circle are shown in Fig. 2(a), and the amplitude temporal curves of the two are shown by the purple curve and the cyan curve, respectively, in Fig. 2(b).

In the measured data, the above amplitude temporal curves are easily affected by the following practical factors. First, besides the shadows produced by moving targets, shadows will also be produced due to the occlusion of buildings (false alarms are caused). Second, the RCS of ground targets will fluctuate due to the change in the viewing angle of Video SAR, which may also bring about the variation of the amplitude temporal curves of pixel points at the corresponding positions. Third, some strong scattering points and the "bright lines" that deviate from the moving targets will also cause variation in the amplitude temporal curves. Fourth, the influence of antenna patterns exists. For example, the pixel points, where the purple circle and the cyan circle are located in Fig. 2(a), are both background areas. However, the purple circle is located on the edge of the scene, and the amplitude curve level is relatively weak. The cyan area is located in the center of the scene, and the amplitude curve level is relatively strong. Fifth, interframe dynamic range fluctuation between Video SAR images may exert influences on the amplitude temporal curves as well.

It can be found from the above analysis that the amplitude temporal curves of some pixel points in the scene are obviously changed due to the movement (leaving or appearing) of the moving targets (Situation 1 and Situation 2). By comparison, the amplitude temporal curves of the background pixel points (Situation 3 and Situation 4 are collectively referred to as "background pixel points") will just fluctuate within a certain range, while the amplitudes of pixel points where moving targets leave or appear will jump irregularly, although they are both affected and interfered by various practical factors. Therefore, the time-series neural network model can be used to learn the characteristics of the amplitude temporal curves

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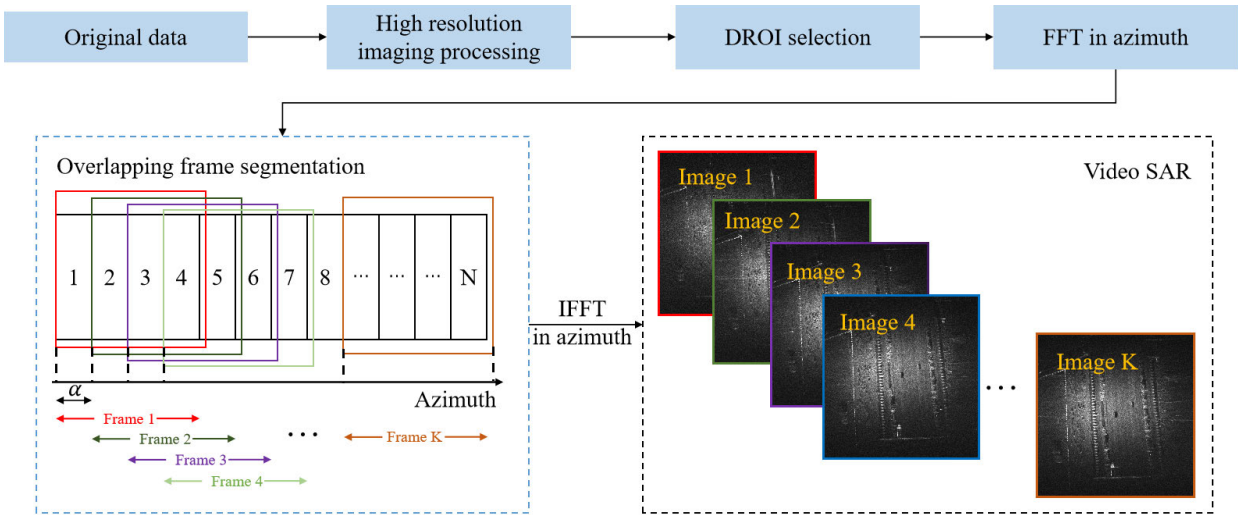



Fig. 3. High-precision registration scheme of Video SAR image sequences.

of different pixel points and classify the pixel points in the scene.

To sum up, the pixel points in the Video SAR scene can be divided into three categories: leave (Situation 1), appear (Situation 2), and background (Situation 3 and Situation 4). On this basis, the pixel points in the scene are classified by using the classified neural network, multiple receptive field fusion neural network model based on frame variation (MRFN-FV); then, the false alarms are suppressed based on the temporal change characteristics of pixel points. Finally, the moving targets in each frame of SAR images are detected and located by using the amplitude-based density clustering algorithm.

## III. Video SAR Echo Preprocessing Method

In order to classify the pixel points in Video SAR images based on the amplitude temporal curves, Video SAR echo needs to be preprocessed first: high-precision registration of Video SAR image sequences and interframe dynamic range correction of Video SAR images.

### *A. High-Precision Registration Scheme of Video SAR Image Sequences*

As the proposed algorithm is based on the amplitude temporal curves of Video SAR, image registration is required for Video SAR interframe images. In this article, we adopt the high-precision registration scheme of Video SAR image sequences based on the full aperture imaging postprocessing algorithm proposed by our research group [26], as shown in Fig. 3. The registration of Video SAR interframe images is conducted during SAR imaging. Assuming that the Video SAR system operates in the spotlight mode, full aperture imaging is conducted for the obtained data, and then, the dynamic ROI (DROI) data are segmented at an overlapping frame rate. The Fourier transform is performed in the azimuth direction to obtain the data of Frame 1, Frame 2, Frame 3, ..., Frame K; the IFFT operation in the azimuth direction is completed for each frame of data to obtain Video SAR image sequences Image 1, Image 2, Image 3, ..., Image K. Then, the video

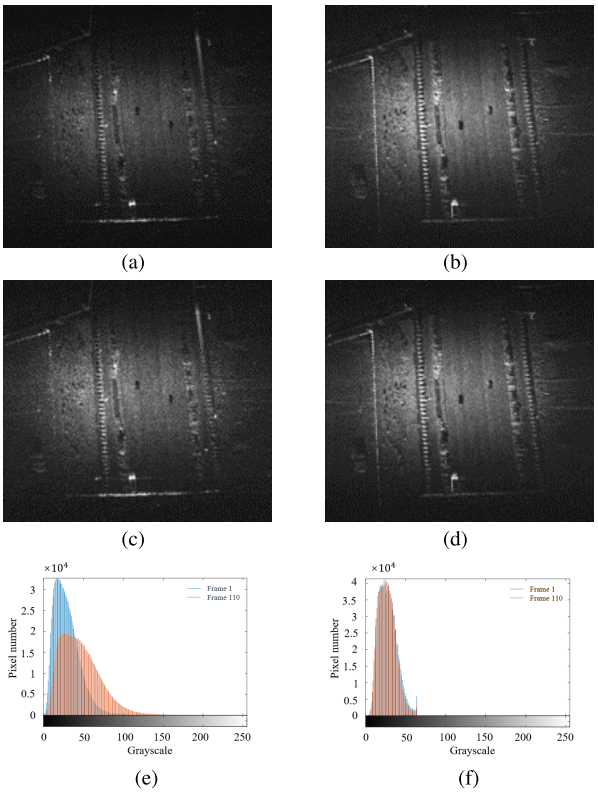



Fig. 4. Correction results of the interframe dynamic range of Video SAR images. (a) Frame 1 (before normalization). (b) Frame 110 (before normalization). (c) Frame 1 (after normalization). (d) Frame 110 (after normalization). (e) Gray histogram (before normalization). (f) Gray histogram (after normalization).

results with corresponding frame rates are obtained through sequential synthesis of the image sequences.

### *B. Interframe Dynamic Range Correction Scheme of Video SAR Images*

There may also be dynamic range fluctuations between the measured Video SAR image sequences in the same scene, as shown in Fig. 4. Fig. 4(a) is the first frame of SAR images, 4(b) is the second frame of SAR images in the same scene,

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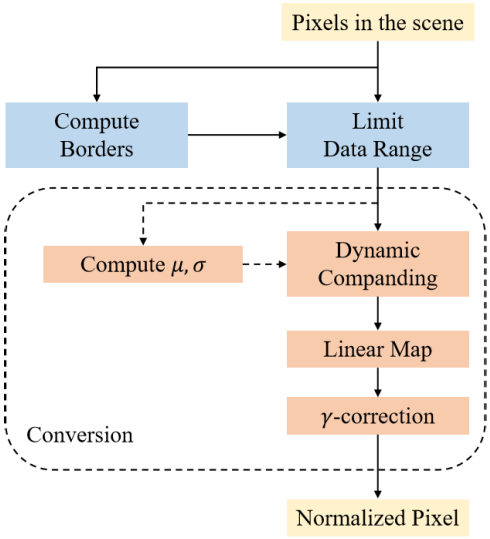



Fig. 5. Conversion algorithm.

and Fig. 4(e) is the gray histogram corresponding to two frames of SAR images. It can be seen that the dynamic range of these two frames of images is obviously different, which will have a great influence on the classification of pixel points in the Video SAR based on the amplitude temporal curves of pixel points. Therefore, it is necessary to correct the interframe dynamic range of Video SAR images.

The specific correction scheme is given as follows. Compute the dynamic range that limits the amplitudes of pixel points, crop the nonzero amplitudes of pixel points to the dynamic range, and then, use the conversion algorithm proposed by Hartmann et al. [27] to process the amplitudes of pixel points. The flowchart of the conversion algorithm is shown in Fig. 5.

The main steps of the conversion algorithm are given as follows. Dynamic companding is conducted, and the amplitudes of pixel points are limited below $\mu + \lambda \cdot \sigma$, where $\mu$ is the average value of statistical pixel points, $\sigma$ is the standard deviation of statistical pixel points, and $\lambda$ is a dynamically changing parameter. The amplitudes after dynamic companding are linearly mapped. The range of amplitudes between the minimum value and the maximum value is mapped to the range from zero to the square of the quantization series so that the data of the lowest amplitude are aligned with zero in the subsequent processing, and the data of the highest amplitude are aligned with the highest quantization level. Then, $\gamma$ correction is conducted for the amplitudes of pixel points to obtain the corrected Video SAR data.

The results after dynamic range correction are shown in Fig. 4(c) and (d), and the corresponding gray histogram is shown in Fig. 4(f). It can be seen that the above method can effectively correct the interframe dynamic range of Video SAR images.

## IV. Video SAR Moving Target Detection, Location, and Tracking Based on the Interframe Amplitude Temporal Curve

The classification of amplitude temporal curves of pixel points by traditional threshold determination methods is easily interfered with by various nonideal factors, which results in inaccurate classification. However, the classified neural network has obvious advantages in learning the characteristics of data changes, so we try to design a classified neural network to learn the characteristics of the amplitude temporal curves of pixel points and classify the pixel points in the Video SAR scene. The overall framework of Video SAR moving target detection, location, and tracking method proposed in this article based on the amplitude temporal curves of pixel points is shown in Fig. 6.

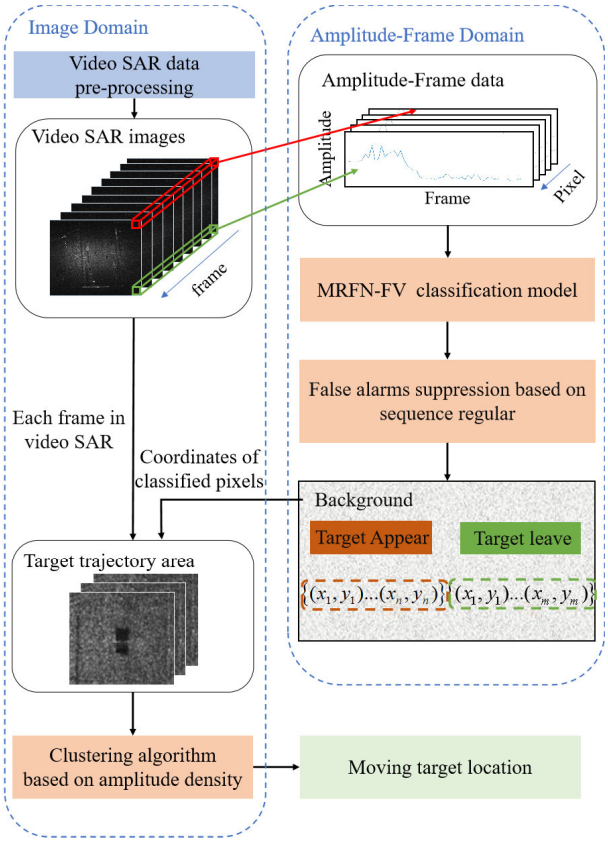



Fig. 6. Overall framework of the Video SAR moving target detection, location, and tracking method based on the amplitude temporal curves of pixel points.

First, after Video SAR echo preprocessing described in Section III, we obtain the amplitude temporal curve of each pixel point in the Video SAR scene by separating the azimuth–range–time 3-D data by taking pixel point as a unit and then input the amplitude temporal curves into the trained classified network (the specially designed MRFN-FV network) to preliminarily classify the pixel points. Second, according to the preliminary classification results, the false alarm suppression of classification results is completed by using the interframe variation characteristics of pixel points, and an improved amplitude-based density clustering algorithm is proposed to locate and track the moving targets.

### A. MRFN-FV Classified Neural Network Structure Model

The commonly used classified network models include CNN [28], [29] and inception [30] network structures. Different from the CNN model, the inception network model can

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increase the adaptability of the neural network structure layer by setting all possible values of network layer parameters, splicing them at the output end of the network, allowing the network to learn the parameters, and combining with the optimal convolution kernel. In practical applications, Video SAR images are easily affected by various nonideal factors, which brings more complex amplitude temporal curves. If CNN and inception neural network structures are directly used to classify pixel points, many false alarms and missing alarms will be generated.

The amplitude temporal curves of pixel points in the scene indicate the overall average change of the amplitudes in the time dimension, while the amplitude temporal variation caused by the background disturbance is unstable. Therefore, the designed classified neural network needs to focus on the overall change of the amplitudes of pixel points during the observation time but ignores the influence of background disturbance. The receptive field (RF) in neural networks plays a crucial role in learning the characteristics of interframe amplitude temporal curves. If the RF is too small, the neural network only learns the local characteristics of the amplitude temporal variation and cannot obviously perceive the overall change level. It is easily interfered with by the background clutter, which may result in the misjudgment of pixel points. Based on the characteristics of Video SAR, this article proposes a new classified neural network: MRFN-FV. This network model fully considers the influence of different RFs on the learning of amplitude temporal curves, which makes the network able to automatically use the information captured by the RFs that are effective for classification. The MRFN-FV network model allows each input data to adaptively adjust the size of its RF according to the multiscale input information. Its inspiration comes from the fact that, when we look at objects of different sizes and different distances, the acceptance region of visual cortex neurons will be adjusted according to stimuli. The MRFN-FV neural network model mainly consists of three parts.

1) *Multiple RFs' Input:* Assuming that the length of the input time sequence is T, it is input into different convolution layers, respectively. The parameters of the convolution layer mainly include the convolution kernel size (represented by *filter_length*), the step size (represented by *stride*), and the number of convolution kernels (represented by *num_filters*). This layer performs an operation of sliding filters of the length $q$ with a stride to 2, while $q = \{5, 10, 15\}$. Then, the T-dimensional time sequences after convolution are input into the inception network structure model, respectively. Compared with the sudden change of the local amplitude of pixel points caused by background disturbance, the amplitude change of pixel points caused by the motion of the moving target is obviously more continuous. The convolution data of multiple input modules are stacked, and then, the weight of the network model is trained by backpropagation.
2) *Inception-Resnet Module:* The basic network structure of the MRFN-FV neural network proposed in this article is an Inception-Resnet network structure. This article applies this neural network to the classification of 1-D amplitude time-series data due to its excellent performance in image classification. An in-depth understanding of the relationship between the amplitude sequence data model and context is one of the keys to ensure correct classification.
3) *Channel Information Statistics:* After learning the features of the input data under different RF selections, global average pooling is used to sort out different channel information, and then, the weight matrix is obtained through the fully connected (FC) layer, the rectified linear unit (ReLU) activation layer, and the batch normalization (BN) layer. The weight matrix is used to weight the feature maps generated by the Inception-Resnet layer, and the channels in the feature maps generated under three kinds of RFs are selected according to the weight information to synthesize the final feature map. Finally, the pixel points are classified by using the FC layer.

The structure model of the MRFN-FV classified neural network, the parameters of the convolution operation, and the relevant dimension of the output feature map after each layer in our experiment are shown in Fig. 7. In Section II-B, we have analyzed four situations and three kinds of amplitude temporal curves of pixel points in the Video SAR images. The pixel points in the scene are divided into three categories in Fig. 7: appear, leave, and background (Background 1 and Background 2).

### *B. False Alarm Suppression Method Based on Temporal Changes of Pixel Points*

The amplitude temporal curves of all pixel points in the scene are input into the MRFN-FV classified neural network to obtain the classification results of three kinds of pixel points. When processing the measured data, the MRFN-FV classified neural network will be affected by various nonideal factors (see the analysis in Section II-B), and false alarms will appear. In particular, the main factors that have a serious effect on the classification results of the MRFN-FV network include the amplitude changes of pixel points caused by "bright lines" and strong scattering points.

According to the different characteristics of temporal changes of pixel points, the false alarm suppression method is proposed in this article, and the flowchart is shown in Fig. 8.

The main content of the false alarm suppression method is divided into the following three parts.

1) *Removal of Single Pixel Point Under Isolation and Misjudgment:* Generally, the Video SAR system has a high resolution, and moving targets often occupy dozens or even hundreds of pixel points instead of isolated single pixel point. The maximum connected domain algorithm can be used to remove the single pixel point under isolation and misjudgment in the detection results of each frame of SAR images. The algorithm is widely used in morphological processing and aims at distinguishing different areas composed of several pixel points with the same pixel value in adjacent regions of the image.

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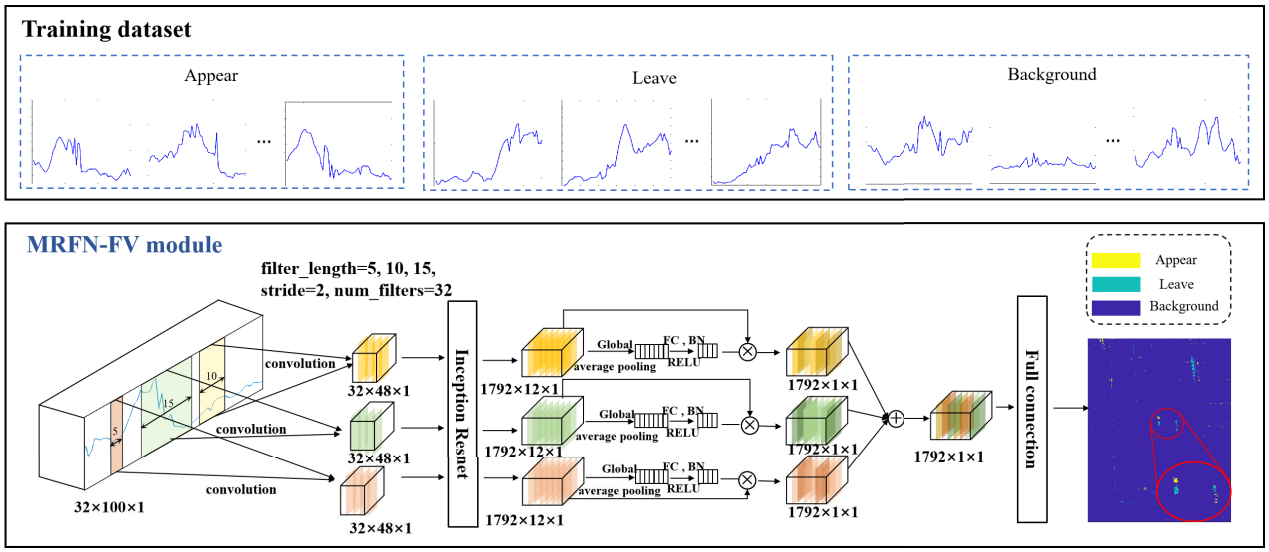



Fig. 7. MRFN-FV classified neural network structure model.

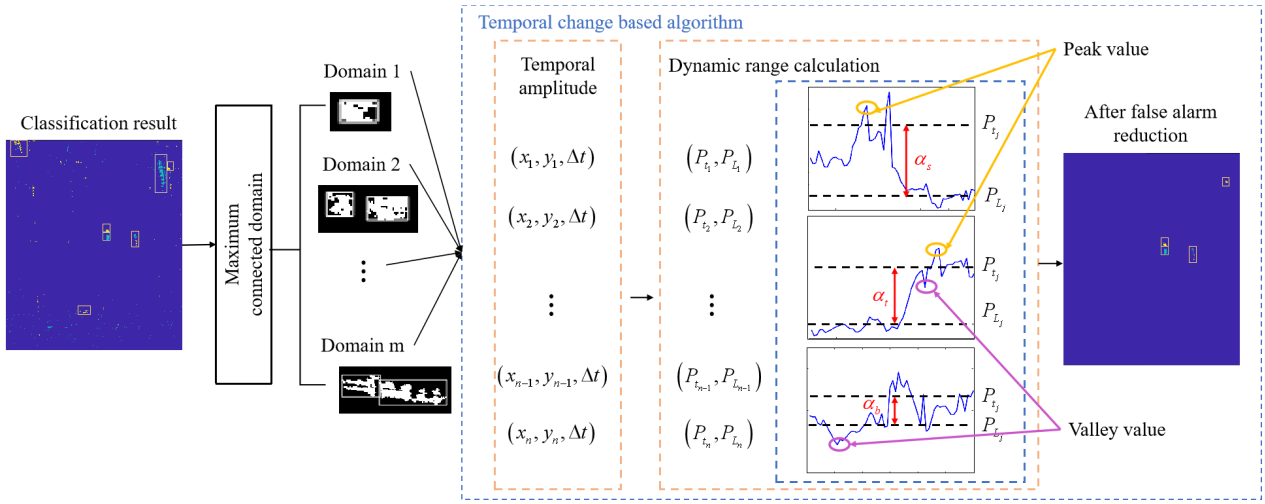



Fig. 8. Flowchart of the false alarm suppression method based on temporal changes of pixel points.

2) *Removal of Single Classification Under Isolation and Misjudgment:* In the classification results of pixel points in the Video SAR images, the pixel points where moving targets appear and leave must emerge in pairs. Therefore, false alarms can be further suppressed by identifying and removing the isolated single classification.
3) *False Alarm Suppression Caused by Movement of Targets and Strong Scattering Points:* In the Video SAR, "bright lines" of moving targets and strong scattering points occupy some areas. Moreover, the amplitudes of the pixel points in the corresponding area also fluctuate greatly, which has similar characteristics to the pixel points where the moving targets appear or leave. It is impossible to remove the classification results under isolation and misjudgment by the maximum connected domain algorithm. The false alarm suppression algorithm based on the characteristics of temporal changes of pixel points is used here to solve the above issues.

The peak value (as indicated by the yellow ellipses in Fig. 8) and the valley value (as indicated by the purple ellipses in Fig. 8) of each amplitude temporal change curve are extracted. As there are many peak values and valley values in the amplitude temporal curves of pixel points, the peak value array $(T_1, T_2, \ldots, T_g)$ and the valley value array $(L_1, L_2, \ldots, L_h)$ can be obtained. Specifically, $g$ and $h$ are the numbers of peak values and valley values in the curves. In the measured Video SAR data, due to the influence of complex background clutter, the amplitude temporal curves will definitely have unstable fluctuations, and the obtained single peak value or valley value cannot reflect the overall state of pixel points during the observation time. Moreover, there exist numerous pixel points in the scene, and each amplitude temporal curve of the pixel point has numerous peak and valley values. To reduce computational complexity, not all values are used for calculation. Therefore, this method sorts the peak value array and the valley value array, respectively, and calculates the average value $P_t$ of before the peak value array

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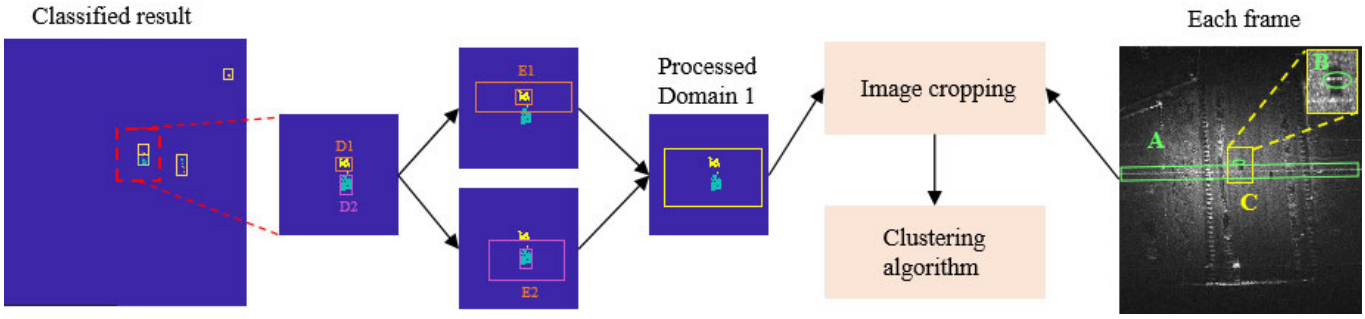



Fig. 9. Schematic of the input processing of clustering algorithm.

$\varepsilon\% - \phi\%$ is taken and the average value $P_L$ of after the valley value array $\varepsilon\% - \phi\%$ is taken. In order to avoid the instability of a single pixel point, this method calculates the average of peak value arrays $\{P_{t_1}, P_{t_2}, \ldots, P_{t_n}\}$ and valley value arrays $\{P_{L_1}, P_{L_2}, \ldots, P_{L_n}\}$ of all the pixel points separately by taking connected domain as a unit, where $n$ is the number of pixel points in the connected domain. In this way, the extremely abnormal peak or valley values caused by background clutter or noise can be significantly excluded, and the computational cost can be relatively reduced. In practice, the values of $\varepsilon$ and $\phi$ are adjusted repeatedly within a small range to obtain superior performance of false alarm suppression. The determination threshold $\delta$ of false alarm suppression can be obtained by the following equation:

$$\delta = \frac{\sum_{i=1}^{n} P_{t_i} - \sum_{j=1}^{n} P_{L_j}}{n}. \tag{1}$$

### C. Moving Target Location and Tracking by the Amplitude-Based Density Clustering Algorithm

In a single-frame image of Video SAR, the shadow feature at the positions of moving targets is an important basis for distinguishing the moving target from the background. However, in the measured Video SAR images, due to the occlusion effects of buildings, shadows will also be left at the corresponding positions. Therefore, if we only rely on the shadow feature to detect Video SAR moving targets, it will bring a lot of false alarms. Meanwhile, the above classification results based on the amplitude temporal curves of pixel points cannot be directly used as the moving target detection results in each frame of Video SAR images.

If the shadow features of moving targets are combined with the classification results based on the amplitude temporal curves of pixel points, can the moving targets in each frame of SAR images be accurately detected and located or not? In view of this, this article proposes the following scheme: crop each frame of Video SAR images based on the position coordinate information of the moving targets in the classification results of pixel points, and then, the amplitude-based density clustering algorithm is used to process the cropped images to detect and locate the moving targets in each frame of Video SAR images.

When analyzing the amplitude change of pixel points in Section III, we find that, if the moving speed of the target is too slow or the moving target is too long, the amplitude change of pixel points in the shadow state for a long time is easily confused with the characteristic of Background 2 in Fig. 2. This is reflected as the interruption of the target trajectory in the classification result figures, which will be determined as two adjacent connected domains after being processed by the maximum connected domain algorithm, as the connected domains $D1$ and $D2$ in Fig. 9. It is impossible to directly

TABLE I
AMPLITUDE-BASED DENSITY CLUSTERING ALGORITHM IN VIDEO SAR IMAGES

**Algorithm:** Amplitude-based density clustering algorithm in Video SAR images

**Input:** The coordinate set of connected domain is $D_i = \{(x_0, y_0, width, length)\}(D_i \in Domain)$, the size of sliding window is $a \times b$, the clustering radius is $R_0$, the minimal number of neighborhood pixel points is *Minp*, and the Video SAR three-dimensional array is $\{(x_1, y_1, \Delta t), (x_2, y_2, \Delta t), ..., (x_n, y_n, \Delta t)\}$.
**Output:** Detection results of moving targets in each frame.
**Steps:** In the obtained classification result figures, the set of the domain where the trajectory is located is recorded as $Domain = \{D_1, D_2, ..., D_n\}$. $n$ is the number of targets, and the total number of frames to be detected is $F$.
**/*Domain traversal**
1.for domain $D_i$ in $Domain$ do:
2.Construct 3D matrix *Ma*, the size of *Ma* is $azimuth \times range \times F$; the value in *Ma* is set to 0;
3.Caculate the center point $[X_0, Y_0]$ of the initial sliding window. This pixel point is taken as the core point;
**/*Classify the pixel points along time axis**
4.for $f$ in $F$ do:
5.Determine whether the core point is the boundary point of other core points. If so, the boundary point will fall into the same category as the core point. If not, a new category will be created.
6.Crop the *f-th* frame of images according to the size of $\eta \times width$ and $\beta \times length$ by taking the core point as the center.
7.Calculate the distance $r_k$ between the amplitude of the pixel point at $[X_0, Y_0]$ and the amplitudes of all other pixel points in the sliding window with a size of $a \times b$ which takes the core point as the center.
8.if $r_k < R_0, r_k \in R$.
9.If the number of pixel points in the set *R* is greater than *Minp*, the central pixel point of the sliding window will be the core point, and all pixel points in the set *R* and their core points will fall into the same category.
10.Sort *R*: the first *Minp* pixel points in the set *R* are categorized as core points, and all the pixel points from the *Minp+1* to the last one in the set *R* are recorded as boundary points.
11.Traverse all the boundary points, and carry out Steps 5-10.
12.Determine whether all the pixel points in the connected domains are traversed. If any pixel points are not traversed, carry out Steps 5-11 for them.

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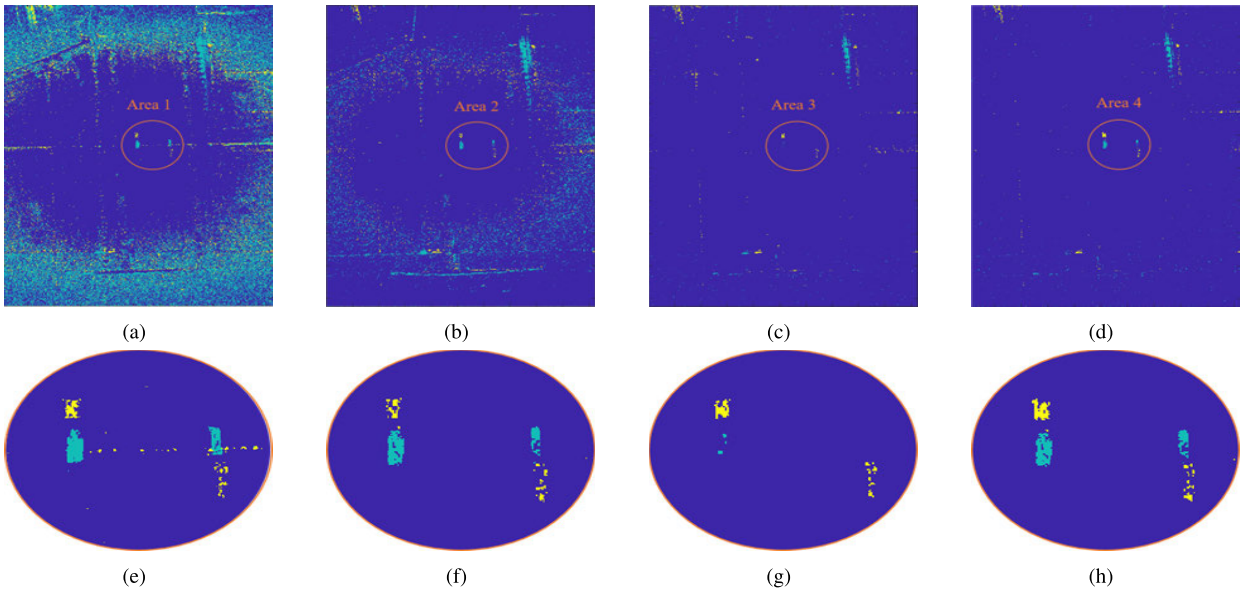



Fig. 10. Comparison figures of detection results of different neural networks. Blue pixel points represent "appear," while yellow pixel points represent "leave." (a) CNN. (b) CNN+LSTM. (c) Inception. (d) MRFN-FV. (e) Area 1. (f) Area 2. (g) Area 3. (h) Area 4.

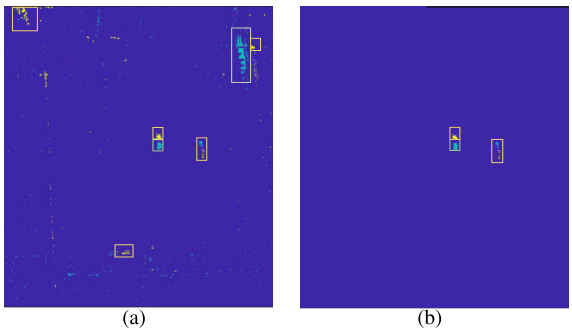



Fig. 11. Classification results of pixel points (a) before and (b) after false alarm suppression.

TABLE II
SELECTION OF PROCESSING PARAMETERS FOR THE MEASURED DATA

| Parameter name | Parameter | Value |
|---|---|---|
| Lower limit of peak(valley) value selection range | $\varepsilon$ | 10% |
| Upper limit of peak(valley) value selection range | $\phi$ | 20% |
| Connected domain width extension factor | $\eta$ | 6 |
| Connected domain length extension factor | $\beta$ | 2 |
| Length of sliding window | a | 400 |
| Width of sliding window | b | 200 |
| Radius of amplitude change | $R_0$ | 0.5 |
| The minimal number of neighborhood pixel points | $Minp$ | 5 |

determine whether it is the trajectory of the same target or two adjacent targets through the classification results. Therefore, it is necessary to process the input of the clustering algorithm, as shown in Fig. 9.

The output of the false alarm suppression algorithm based on temporal changes is the coordinate of the largest connected domain. Assuming that multiple connected domains are labeled as $Domain = \{D_1, D_2, \ldots D_n\}$, $n$ is the number of connected domains in the scene. $P = \{p_1, p_2, \ldots, p_m\}$ is the set of pixel points in the domain $D_i$, and $m$ is the number of pixel points in the set. Based on the connected domains, the width and length of each connected domain are, respectively, extended to $\eta$ and $\beta$ times its original width and length, with the connected domains $E1$ and $E2$ shown in Fig. 9. If there is an intersection between the areas of the two connected domains, the two connected domains are incorporated into a connected domain to obtain the *proposed Domain 1*, and then, each frame of Video SAR images is cropped according to the extended connected domains. In practice, the selections of $\eta$ and $\beta$ are mainly based on the scale of the moving target's trajectory. From Fig. 9, we can see that two adjacent targets are very close, so the length extension factor $\beta$ should not be too big so as to determine whether they are the trajectory of a moving target or not. Moreover, the target only occupies a relatively small area in the whole Video SAR image. Theoretically, the selection of width extension factor $\eta$ should meet such a requirement that the width of the *proposed Domain* can cover the whole range of moving target trajectory, and the proper proportion of the target area in the whole image needs to be considered as well. Finally, the set of pixel points in the cropped images is used as the input of the clustering algorithm. The input of this method can effectively avoid the interference of unnecessary noise in the background.

In the amplitude-based density clustering algorithm, we take the central point of the connected domain as the core point and set the coordinate of the central pixel point of a connected domain as $[X_0, Y_0]$

$$X_0 = x_0 + \frac{1}{2} \cdot \text{width} \tag{2}$$

$$Y_0 = y_0 + \frac{1}{2} \cdot \text{length} \tag{3}$$

where $(x_0, y_0)$ is the coordinate of the upper left point of the connected domain, the $width$ refers to the width of the connected domain, and the $length$ refers to the length of the connected domain. We set a sliding window with a size of

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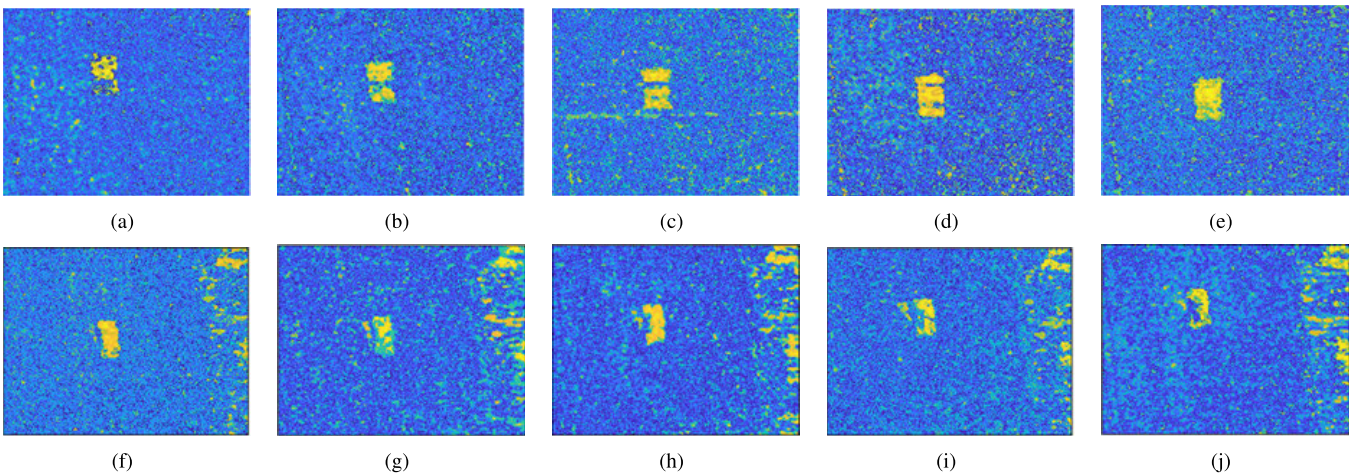



Fig. 12. Results of single target detection by the DBSCAN clustering algorithm. (a)–(e) Domain 1 in Frames 1, 10, 20, 30, and 40. (f)–(j) Domain 1 in Frames 60, 70, 80, 90, and 110.

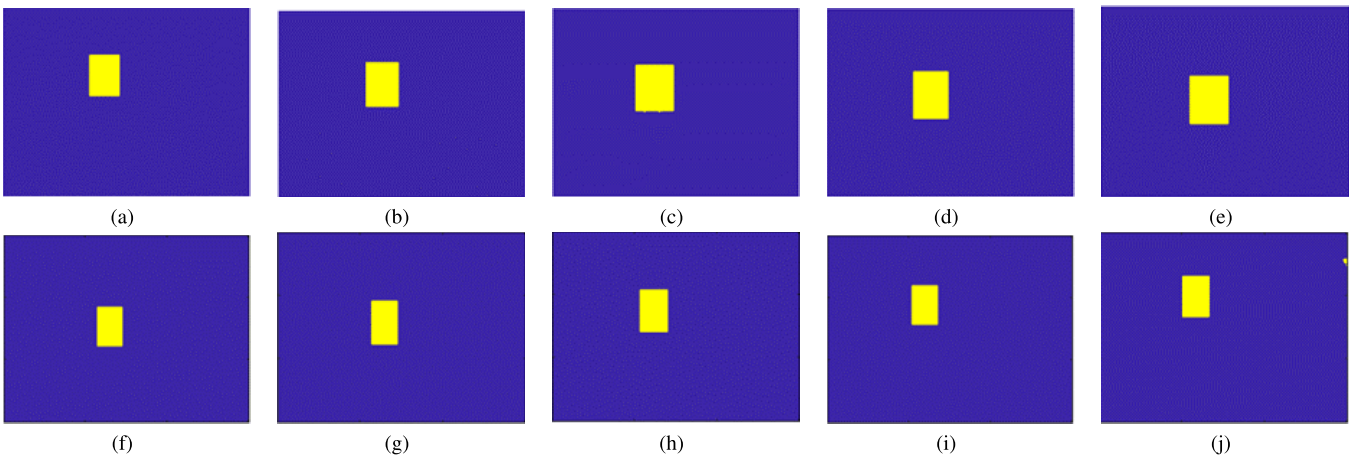



Fig. 13. Results of single target detection by the amplitude-based density clustering algorithm. (a)–(e) Domain 1 in Frames 1, 10, 20, 30, and 40. (f)–(j) Domain 1 in Frames 60, 70, 80, 90, and 110.

$a \times b$ by taking the amplitude of the pixel point at $[X_0, Y_0]$ as the center and calculate the distance $r_k (k \in (1, m))$ between the amplitudes of all other pixel points in the sliding window and the amplitude of the central point. The size of the sliding window is equal to the *proposed Domain* obtained by input preprocessing steps as described previously. If $r_k < R_0$ ($R_0$ is the radius of amplitude change), $r_k$ will be included in the set $R$. If the number in $R$ is greater than the set minimal number of neighborhood pixel points (*Minp*), the central point will be the core point, or it will be the noise point. The elements $r_k$ in set $R$ are sorted in ascending order, the pixel points from *Minp* to the last one will be the boundary points, and the boundary point will be the center point of the next sliding window. By traversing all the pixel points in the whole neighborhood, the amplitudes of pixel points in the scene are clustered, and then, the moving targets in each frame of SAR images are detected. It is worth mentioning that, in the subsequent experiment, the range of $R_0$ and *Minp* is roughly estimated and then adjusted repeatedly to achieve optimal performance of the clustering algorithm.

When the amplitude-based density clustering algorithm is used to process each frame of images, the "bright lines" of some moving targets may appear at the shadow position, which will lead to the interruption of shadows, with Area B shown in Fig. 9. At this moment, the coordinate of the extended connected domain can be used as a reference. If the coordinate of the interrupted position is within the extended connected domain, the pixel points divided into two categories will fall into the same category, and the pixel points at the interrupted positions will also fall into this category.

The flowchart of the proposed amplitude-based density clustering algorithm is summarized in Table I.

## V. Experiment

We use the measured Video SAR data to verify the effectiveness of the proposed method. The measured data are collected by the Video SAR system operating in the spotlight mode, as described in Section II. In the processing of the measured data, the selection of relevant parameters is shown in Table II.

### A. Construction of Training Set

By processing the data obtained from a real Video SAR system, we obtain a total of 110 frames of measured data images in the same scene, and the training set is constructed based on the interframe amplitude temporal curves of some

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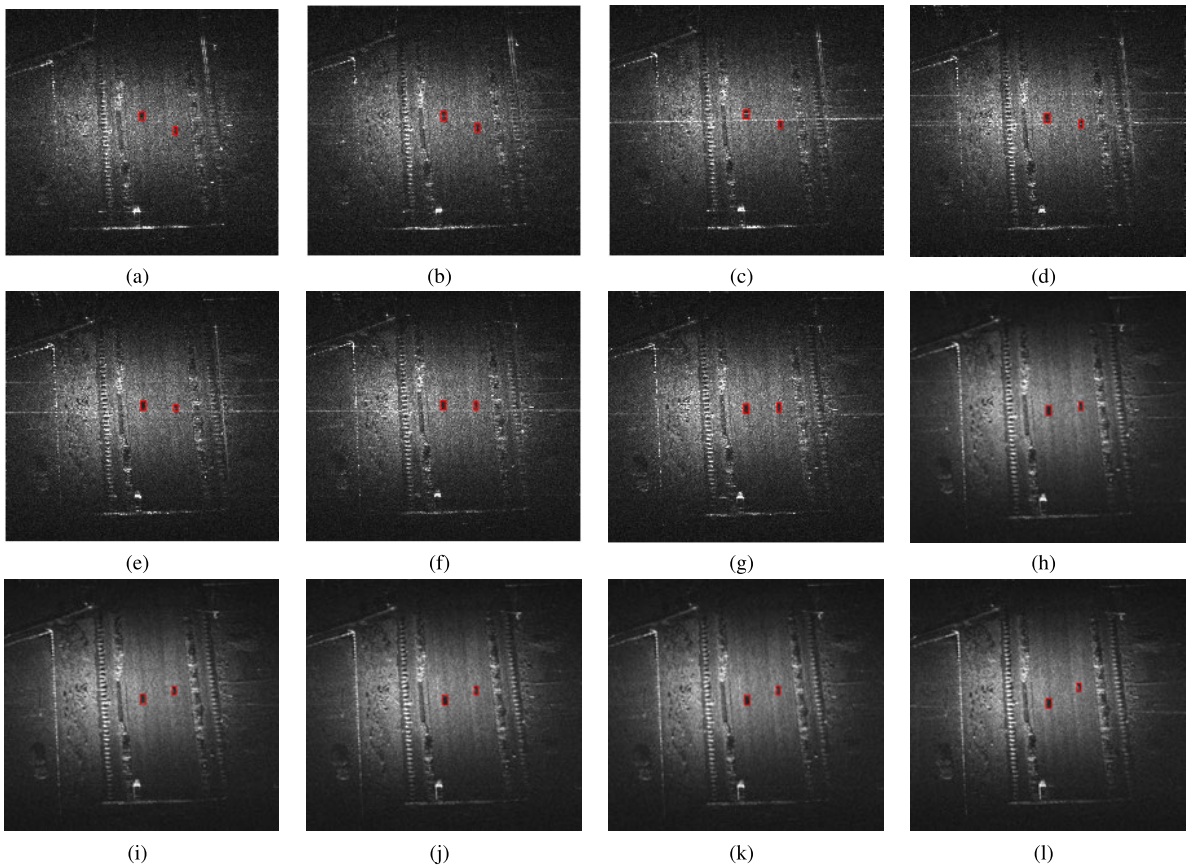



Fig. 14. Target detection results of some frames of Video SAR. (a) Frame 1. (b) Frame 10. (c) Frame 20. (d) Frame 30. (e) Frame 40. (f) Frame 50. (g) Frame 60. (h) Frame 70. (i) Frame 80. (j) Frame 90. (k) Frame 100. (l) Frame 110.

pixel points in the images. The training set consists of three kinds of data: the pixel point dataset where the moving target appears, the pixel point dataset where the moving target leaves, and the background dataset. The pixel point scale of the experimental scenario is $2048 \times 1849$. In the constructed training sample set, the number of pixel points where the moving target appears is 1366, the number of pixel points where the moving target leaves is 1037, and the number of background pixel points is 1883.

### *B. Classification Results of Pixel Points Based on Different Neural Networks*

In order to verify the effectiveness of the proposed MRFN-FV neural network, MRFN-FV is compared with CNN, CNN-LSTM, and Inception in the experiment. That is, the training data in section V-A are input into CNN, CNN-LSTM, Inception, and MRFN-FV neural network models for training, and the obtained detector is used to classify pixel points in the Video SAR. The final classification results of pixel points in the scene are shown in Fig. 10. Specifically, the blue pixel points represent "appear," and the yellow pixel points represent "leave."

Fig. 10(a) and (e) shows the classification results of pixel points based on the CNN neural network. The neural network based on CNN has a simple structure and fewer convolution layers, and the learning of characteristics of pixel point variation is relatively rough. The simple disturbance of the SAR background will cause misjudgment of pixel points, so the classification effect based on the CNN network is relatively poor.

Considering that the pixel points to be classified are the temporal data and the excellent performance of LSTM [31] neural network can be achieved in processing the temporal correlation between data, the CNN+LSTM neural network structure model is selected to classify the above datasets, and the results are shown in Fig. 10(b) and (f). From the classification results of pixel points in the Video SAR, it can be seen that, with the increase in network layers and considering the correlation between data on the time axis, false alarms are greatly suppressed. However, from the classification results of pixel points in the whole scene, the classification effect based on the CNN+LSTM neural network is still not ideal.

Fig. 10(c) and (g) shows the classification results of pixel points in the Video SAR based on the inception neural network. Inception is a typical classified neural network. It can be seen from the results that false alarms are further reduced, but the detection effects of pixel points where moving targets appear and leave are reduced.

Fig. 10(d) and (h) shows the classification results of pixel points in the Video SAR based on the proposed MRFN-FV neural network. It can be seen that, compared with the previous

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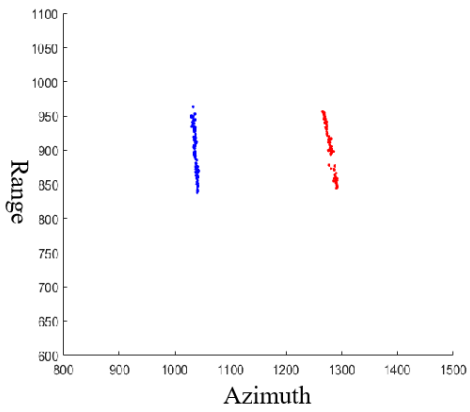



Fig. 15. Trajectory figure of moving targets of 110 frames.

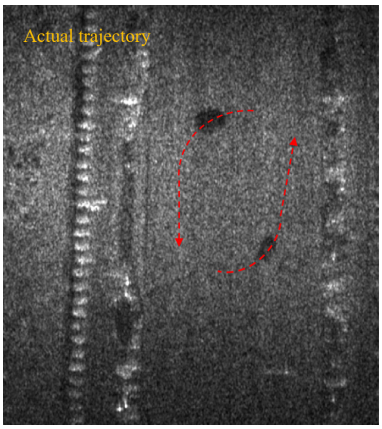



Fig. 16. Trajectory of moving targets in a turning condition.

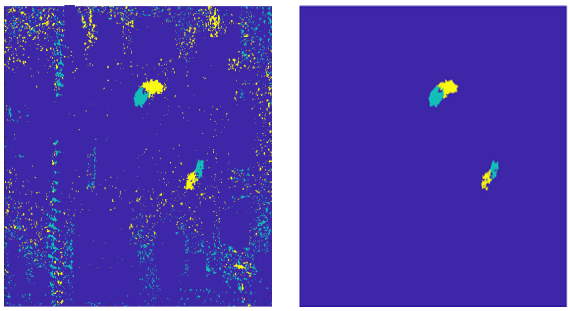
three neural networks, the classification effect based on the proposed MRFN-FV neural network is the best.

### *C. Results of False Alarm Suppression Method Based on Temporal Changes of Pixel Points*

After classifying the pixel points in the Video SAR images based on the MRFN-FV neural network, the maximum connected domain algorithm is applied to obtain the preliminary results of false alarm suppression. The discrete pixel points distributed in the background can be eliminated significantly, and several areas that are potentially moving targets are reserved and labeled as yellow rectangles in Fig. 11(a). On this basis, false alarms are continuously suppressed by using the method described in Section IV-B, and the final suppression results of false alarms are shown in Fig. 11(b). It can be seen from Fig. 11 that the proposed false alarm suppression method can effectively remove the false alarms beyond "appear" and "leave."

### *D. Results of Moving Target Location by the Amplitude-Based Density Clustering Algorithm*

The positions of "appear" and "leave" are not completely the same as the actual positions of moving targets. When processing the measured data, the approximate range of the actual positions of moving targets is determined in each frame of SAR images according to the positions of "appear" and "leave" so as to realize a rough location, and then, the moving targets can be accurately located and tracked by using the amplitude-based density clustering algorithm proposed in Section IV-C.



(a) (b)

Fig. 17. Classification results of pixel points (a) before and (b) after false alarm suppression.

The final results of moving target location by inputting the segmented images to the traditional DBSCAN algorithm [32] and the proposed amplitude-based density clustering algorithm are shown in Figs. 12 and 13, respectively. In Fig. 12, the traditional DBSCAN clustering algorithm only conducts clustering according to the distance of the amplitudes of pixel points. In the final result, the pixel points in the 2-D rectangular box where the target is located are marked in the same category (the yellow area in Fig. 12), while some background pixel points and single-scattering-pixel points displayed as yellow points are also classified as the same category. This will result in the misjudgment of moving target positions. After analyzing the clustering results in Fig. 12, it is found that the shadows of moving targets in some frames will be interrupted by their bright lines or background clutter [see Fig. 12(b) and (c)]. However, the amplitude-based density clustering algorithm can remove the scattered and misjudged pixel points in the SAR background, and it can judge whether two continuous areas are the same target after segmentation.

Some detection results of the positions of the moving targets in each frame are shown in Fig. 14. Taking the Barycentric coordinates of the shadows in the detection boxes as the target positions, the target positions in the 2-D scene (the azimuth–range direction) are drawn to obtain the trajectory figures of the moving targets in 110 frames of images, as shown in Fig. 15.

### *E. Comparative Detection Results of Different Test Datasets*

The neural network based on the shadow features of moving targets can detect the actual positions of the moving targets in the Video SAR images. In practical applications, the shadow features of moving targets in the Video SAR images are diverse (especially when the moving targets turn round or the moving targets are occluded by the bright lines), but there is a small amount of the measured data, so it is difficult to include all the possible shadow shapes of moving targets in the training sample set. The moving target detection method based on the amplitude temporal curves of pixel points in the Video SAR proposed in this article can solve the above problems well.

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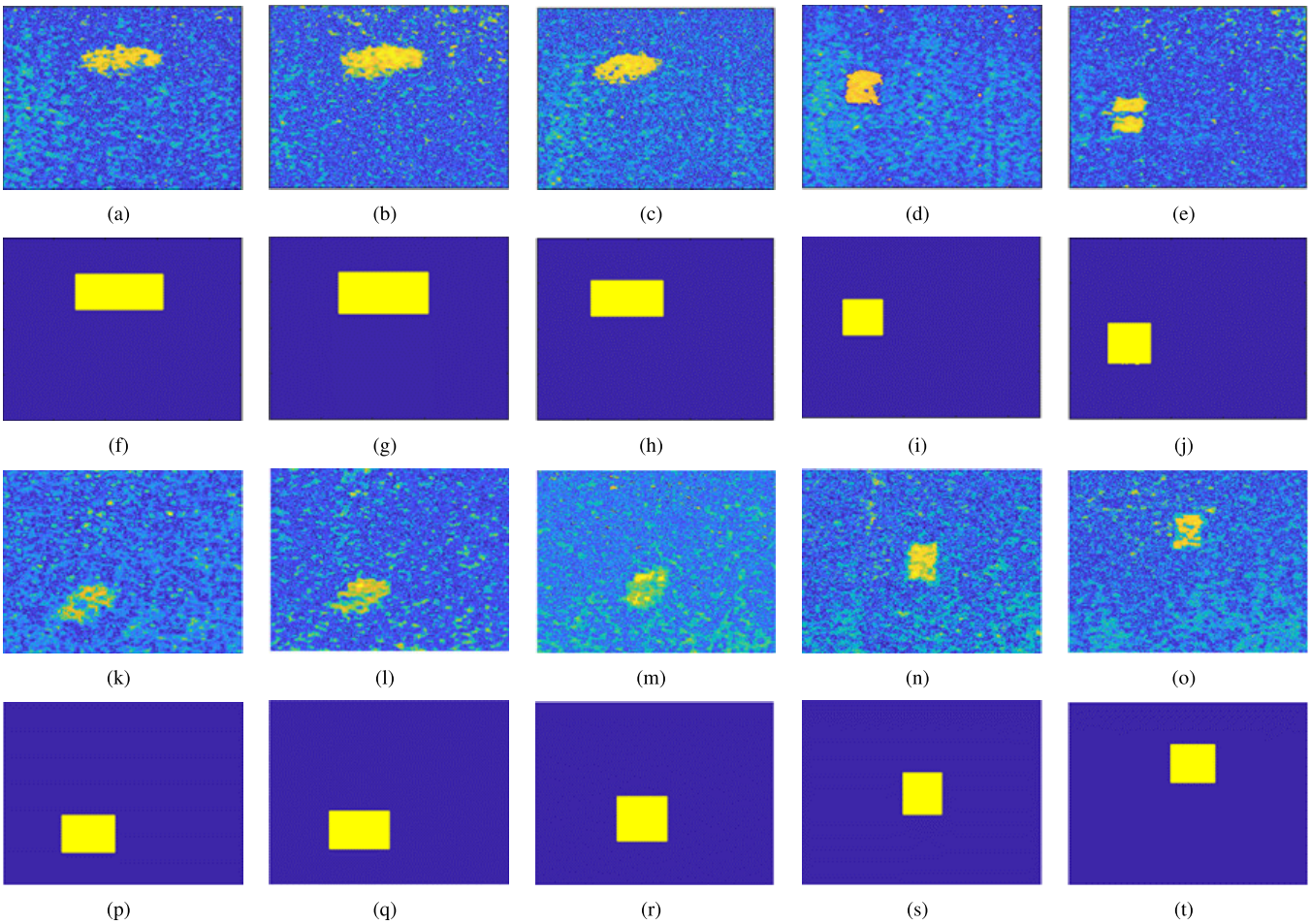



Fig. 18. Comparison figures of some results of clustering algorithms. (a)–(e) Domain 1 in Frames 31, 36, 55, 100, and 150 by the DBSCAN clustering algorithm. (f)–(j) Domain 1 in Frames 31, 36, 55, 100, and 150 by the amplitude-based density clustering algorithm. (k)–(o) Domain 2 in Frames 31, 36, 55, 100, and 150 by the DBSCAN clustering algorithm. (p)–(t) Domain 2 in Frames 31, 36, 55, 100, and 150 by the amplitude-based density clustering algorithm.

Another test dataset is input into the neural network model for comparative experiment analysis. This test dataset includes not only the straight movement of the moving targets but also the turning of the moving targets, as shown in Fig. 16. When the moving target makes a turn, the shape of the shadow is no longer a rectangle. When using the neural network to intelligently detect moving targets in a turning condition based on the shadow features, the training dataset is necessary to cover the moving target samples in a turning condition.

In order to further verify the effectiveness of moving target detection based on the amplitude temporal curves of pixel points, the quantitative comparison in terms of Faster-RCNN [33], DETR [34], Yolov5 [35], and the proposed method is performed based on the measured Video SAR data in Table III, and the experimental results are compared and analyzed accordingly. We still use 110 frames of measured data images described in Section V-A as the training sample sets of the four neural network models, and the test set is changed into a dataset containing two moving targets in a turning condition in the same test scene. There are totally 211 images in the test set, and there are two moving surveillance targets in each image. That is, there are 422 targets to be detected. The training sample dataset is separately fed into Faster-RCNN, DETR, Yolov5, and MRFN-FV for training, and the obtained training models are then utilized to detect the images in the test dataset. The preliminary detection results of the trained Faster-RCNN, DETR, Yolov5, and MRFN-FV network are presented in Table III, which contains 358, 392, 408, and 422 correctly detected targets, 0, 22, 0, and 4 false alarms, and 64, 30, 14, and 0 missing alarms, respectively (the confidence level is set as 0.6). Faster-RCNN is a classical two-stage algorithm, which is widely used in object detection and classification tasks. It first generates a series of anchors, which may contain targets to be detected by using the region proposal network (RPN) based on feature extraction. Then, prediction boxes will be obtained by filtering the region proposals and several postprocessing steps. However, due to the monotonous

TABLE III
COMPARISONS OF DETECTION PERFORMANCE ON THE MEASURED DATA BASED ON DIFFERENT METHODS

| Methods | Correctly detected targets | False alarms | Missing alarms | Training time | Inference time |
|---|---|---|---|---|---|
| Faster-RCNN | 358 | 0 | 64 | 32284s | 0.52s |
| DETR | 392 | 22 | 30 | 34422s | 1.72s |
| Yolov5 | 408 | 0 | 14 | 14033s | 0.26s |
| MRFN-FV | 422 | 4 | 0 | 548.31s | 0.37s |

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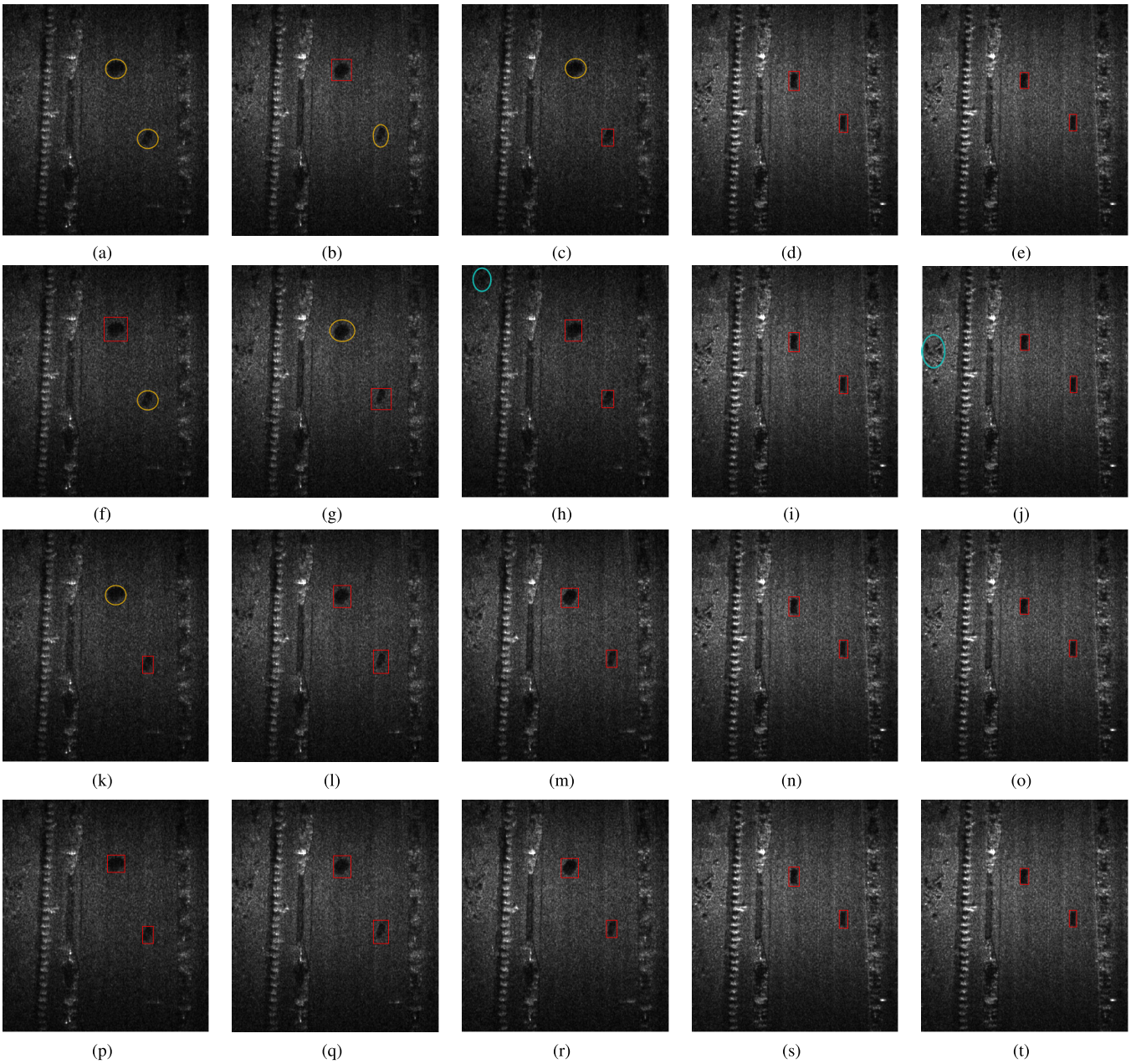



Fig. 19. Results of moving target location by Faster-RCNN, DETR, Yolov5, and MRFV-FV (red rectangle represents correctly detected target, yellow ellipse represents missing alarm, and cyan ellipse represents false alarm). (a)–(e) Results of moving target location by Faster-RCNN. (f)–(j) Results of moving target location by DETR. (k)–(o) Results of moving target location by Yolov5. (p)–(t) Results of moving target location by MRFV-FV.

color of Video SAR images and the poor behavior of Faster RCNN in detecting minor targets, quite a few missing alarms emerge in the detection results compared with the other methods. Comparatively, as a representative of a one-stage object detection network, Yolov5 views the object detection task as a regression problem and directly inputs the whole image into the network to infer by exploiting global features. Therefore, the network will be more flexible in detecting targets with multiscale and perform better in the detection of minor targets. The missing alarms are significantly reduced, and the number of correctly detected targets increases accordingly in the detection results. Different from traditional object detection algorithms, DETR removes several postprocessing steps and is not tied to prior knowledge, which directly simplifies object detection as a set prediction task. It adopts an encoder–decoder architecture and operates more efficiently since it predicts all objects at once and performs a parallel working way. Thus, decent prediction results can be achieved. However, the false alarms of DETR are relatively unsatisfactory compared with other networks due to the difficulty in distinguishing moving targets from dark backgrounds and lack of training samples. The MRFN-FV network fully utilizes the temporal change characteristics of pixel points in Video SAR images instead of 2-D morphological features, which makes it easier to obtain abundant training samples and achieve better performance of moving target detection and location. Therefore, the moving targets in the scene can be completely detected with acceptable false alarms compared with the other three networks.

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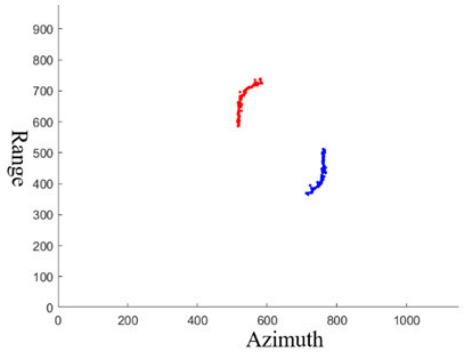



Fig. 20. 2-D trajectory figure of moving targets obtained by the proposed algorithm.

The training and inference times of the four neural networks shown in Table III demonstrate the effectiveness of the proposed method further. The training process consists of 200 epochs, the experiment platform is NVIDIA GeForce RTX 3060 Ti GPU, and the deep learning framework is Pytorch (version 1.7.1). The network structure of Faster-RCNN is more complex than Yolov5, and consequently, the training time of Faster-RCNN is much longer, probably 2.3 times that of Yolov5. In addition, the inference speed of a single image of Faster-RCNN is also quite slow, but the difference is not significant. The inference time of Yolov5 is the shortest since it uses a lightweight network. DETR adopts a transformer structure and contains multiple encoder–decoder layers, which results in its very long training time and inference time. Due to the adoption of time sequence data for training and testing, the data volume of MRFN-FV is much smaller, so the operation time is only 548.31 s and the inference time is also satisfactory even if more epochs are trained.

In particular, the classification results of pixel points obtained by the MRFN-FV classified network before false alarm suppression are shown in Fig. 17(a). The results of false alarm suppression based on the characteristics of interframe temporal changes of pixel points are shown in Fig. 17(b).

As shown in Fig. 17(b), some false alarms can be effectively suppressed according to the temporal change characteristics of pixel points, and the preliminary location results of the moving targets can be achieved. Then, the amplitude-based density clustering algorithm is used to cluster each frame of images and obtain the results in Fig. 18. In particular, Fig. 18(a)–(e) and (k)–(o) shows the results of inputting cropped images into the DBSCAN clustering algorithm, and Fig. 18(f)–(j) and (p)–(t) shows the results obtained by the amplitude-based density clustering algorithm. It can be seen that the classification situation is complicated in the results obtained by the DBSCAN clustering algorithm, and there are many misjudged pixel points, while the moving targets can effectively be located and detected by the amplitude-based density clustering algorithm.

The final location results of moving targets obtained by using Faster-RCNN, DETR, Yolov5, and MRFN-FV are partially shown in Fig. 19. As shown in Fig. 19(a), (g), and (k), the targets at the turning point cannot be detected by Faster-RCNN, DTER, and Yolov5 due to insufficient training samples, but it can be accurately detected by MRFN-FV. In addition, compared with other networks, several false alarms emerge in the detection results of DETR due to the misjudgment of the dark blocks in the background whose shapes are similar to the shadows of moving targets, which can be partly mitigated by setting higher confidence of targets. On the basis of detection results produced by MRFN-FV, we calculate the Barycentric coordinates of the targets in each frame to obtain the motion trajectory of the targets in Fig. 20.

## VI. Conclusion

Based on the in-depth analysis of the measured data of the Video SAR system and combining it with the shadow features of moving targets, this article proposes a method of Video SAR moving target detection, location, and tracking based on the interframe amplitude change curves of pixel points in the Video SAR. Pixel points in the video SAR images are divided into three categories: appear, leave, and background. The primary classification of pixel points in the Video SAR is realized by the specially designed classified neural network MRFN-FV. The false alarm suppression method based on the characteristics of interframe temporal change of pixel points is proposed to further reduce the false alarms, and the amplitude-based density clustering method is presented to accurately locate and track the moving targets in each frame of SAR images. Finally, the effectiveness and universality of the proposed method are verified by the measured data of a real Video SAR system. The proposed method can robustly detect and track the moving target despite the size of the scene, and the operating speed is also considerable when compared with the other algorithms mentioned in this article.

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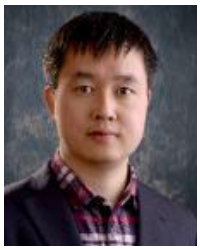
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